

Transcription & Transkribus editing guidelines (SLang, 2025-09)

Efficiency / strategy

To estimate how long you will actually take, try a few pages and time yourself. The first pages will naturally take longer, but once familiar with the workflow and relevant Transkribus features, one page should take around 5–10 minutes (with the Michael Maier texts), especially when using LLM suggestions for correction (since the *Michael Maier* texts contain relatively few errors).

Work strategically in batches: complete all layout tasks first, then correct the text. This reduces unnecessary clicking.

Please track your times, and if some features are not feasible, list what you can or cannot implement so I can prioritise.

I have written a blog post suggesting an efficient workflow for correcting OCR ground truth (<https://latex-ninja.com/2025/06/01/how-to-create-training-ground-truth-data-to-improve-ocr-htr/>)
→ key takeaways:

- Do not attempt linear reading, as it is slower and makes errors harder to spot.
- Instead, focus on specific features, one at a time (for all instances on the page, then move on to the next thing).
 - For example, line endings often contain mistakes such as missing punctuation. According to our guidelines, when a word is broken at a line break, a logical negation sign should be used as a hyphen. Sometimes this is correct in the transcription, but often it is not. It can therefore help to check all line endings first.
 - For special characters, copy the symbol to your clipboard and correct all instances systematically.
 - I recommend creating or using/learning existing keyboard shortcuts for frequently used functions, as this is much faster in the long run than repeated manual clicking (despite the initial learning curve).

File format and export

As discussed, the contract mentions TEI-XML-compatible format in Transkribus, eScriptorium, or whichever tool you prefer. This means that the visual characteristics of the layout are represented in these tools in such a way that the automatic platform export to TEI-XML (available to me as a Pro user, probably not to you) contains the relevant layout features.

Principle: Final text output should “function” without the images (i.e. retain layout information and produce correct reading order)

The goal is to create a (machine) readable text in correct reading order, true to what is visible, (as) efficient (as possible) to produce, and consistent with the partially normalised transcription format of the *NOSCEMUS* Transkribus model.

An overarching guideline is efficiency: preserve what is there, do not spend too much time harmonising minor details (I’d rather you used that time to annotate meaningful layout features such as headers, page numbers or catch-words), and avoid introducing interpretations. **The focus is on producing a usable edition faithful to the text’s semantic content and layout.**

However, please also note that expected correctness standards for digital scholarly editions are practically zero (character-level!) mistakes in the transcription (although perfection, of course, is never 100% possible).

Another guiding principle is that **if the image were removed, the transcription output should still make sense**. Occasionally, layout segmentation errors will break the reading order. These should be corrected.

- For example, if a page has two columns, they should be encoded and ordered correctly to preserve the reading order. You can adjust the layout and line order automatically by enabling the layout section. Where possible, annotate page numbers, headers, or catchwords using structural tags with regions. These may need to be activated in the Transkribus configuration.
- Transkribus offers an option to auto-renumber lines. To correct line order, go to the text tab, enable layout editing, and drag lines or reorder them. Under *Order lines by column* (with the already selected left-to-right ordering). If the current order is correct, avoid reordering, as it can overwrite good data. Use such “replace all (instances on a page)” features cautiously: they may overwrite correct things and your manual work if applied at a later stage in the correction process. So if you want to use them, better do it before investing lots of work into the manual correction of the page in question.
- Italic text should be tagged as such, using the tagging functionality.

LLM-powered OCR-postcorrection workflow

As discussed, I have also provided an LLM-powered OCR-postcorrection workflow:

<https://github.com/sarahalang/LLM-powered-OCR-correction/blob/main/DHd2025-workshop-sy>

[mbole-postcorrection/magnet-p721-korrigiert.jpg](#)

In my experience (as described in the repo), GPT-3.5 Turbo has performed better than the “more intelligent” models (like 4o). It is inexpensive and easy to set up if you follow the directory structure described in the repo.

In my experience GPT-3.5 Turbo works better than other so-called more intelligent models, is inexpensive, and easy to set up if you follow the directory structure.

To obtain new TXT files and images from Transkribus, use the **export** function in the collection overview. Go into the desired file, then select all pages, and under *Actions* choose *Export*. Download “one TXT file per image” and the corresponding images, place them in the correct / indicated directory structure, and then run the script (korr-ocr.py). The *generate diff* script will automatically create relevant **git diff** commands.

eScriptorium also offers this functionality, so you may migrate there if you prefer. You may need to export existing transcriptions from Transkribus in **page.xml**. Documentation should provide further details (<https://escriptorium.readthedocs.io/en/latest/transcribe/>), see also OCR-BW who might offer you access to their eScriptorium instance: <https://ocr-bw.bib.uni-mannheim.de/>

Page Layout and Segmentation Errors

As for layout, it would be highly desirable to annotate features such as headers, catchwords (“Reklamante” at the bottom of pages which is necessary for book binding/organisation rather than semantically meaningful for the text), page numbers, and headings.

For region-level annotations, use **catchword**, **marginalia**, **page number**, **signature mark**, **header**, **heading**. “Footer” is unnecessary if catchwords or other more specific tags are used. Within the text, italic tagging is particularly important. For marginalia, ensure they are inserted so the reading order remains correct.

Note that “heading” refers to titles within the text, while “header” is the top-of-page element (such as containing a running title).

Italics should also be tagged. Unlike structural tags (e.g. headers, marginalia) which appear with the regions in Transkribus, italic is a word-level tag applied in *tagging mode*. Once tagging mode is enabled, highlight the word and select italics from the pop-up menu, or use the shortcut (Cmd + I).

If single lines are erroneously broken up, link them back together.

Ornaments, Images, Drop Capitals

In *Michael Maier* texts, OCR sometimes produces **junk text from ornaments**. The fastest way to remove this is to overwrite it with an image region, which automatically deletes the false text.

If an actual image, decorative initial (drop capital) or ornament occurs, annotate it as such, though these are rather rare.

The above method of tagging a region as image (thus auto-removing related text lines) can also be used to quickly remove digitisation-related info (such as digitised by Google or rights disclaimers at the start of documents).

For metadata pages, retain relevant details such as publication year. Delete disclaimers (e.g. “digitised by Google”) quickly by converting them to an image region and deleting the region.

Marginalia (esp. For Maier texts) and Columns

Erroneously merged lines: If marginalia or separate columns are merged into one line, split them.

This can usually be done by simply creating a new region around marginalia; Transkribus should now automatically re-assign such regions and split lines, though they may still be in incorrect order.

In Transkribus with standard settings, marginalia are placed at the end of the page, which undermines their function.

In the *Michael Maier* texts, where marginalia appear, this can be improved by splitting the main text region in two / multiple parts.

If done indiscriminately, this may break up sentences.

Ideally, you could use a custom polygon line split at the sentence level: draw a split that follows the top line of the margin note, so the sentence is preserved. This ensures correct placement without disrupting the text flow, provided this additional step is feasible within time constraints.

For splitting regions efficiently, create a large region or use the existing one, then apply the **S** shortcut to perform a custom line split (e.g. dividing two columns). Take care, as incorrect column splits may break reading order and can be time-consuming to fix.

Guidelines for the partially normalised transcription following NOSCEMUS GM conventions

The edition should follow a partially normalised format rather than a fully diplomatic one.

If text is truly unclear, retain a diplomatic rendering rather than guessing (and maybe inventing something incorrect that simply isn't there on the page).

If confident in the reading, especially with Latin knowledge, corrections may be applied.

Do not add enrichment (e.g. named entities like names, dates, places); this will be handled later with semi-automatic tools. The main priority is to preserve relevant aspects of the visual layout, so that the output text still functions in all of its aspects if the image is removed.

If some pages' existing machine transcriptions are totally unusable, I can rerun recognition with my Pro account. Please notify me. Rerunning recognition will delete your progress, including any segmentation, depending on settings. If rerunning a model, use the *NOSCEMUS General Model (GM) 6* and enable the language model in advanced settings. Settings make a considerable difference; incorrect ones can produce much worse results.

The transcription guidelines are based on the *NOSCEMUS General Model*. Although they are not explicitly published, I know them partly from discussions with the creator, Stefan Zathammer, and from a joint publication

(<https://library-publishing.uni-graz.at/index.php/lp/catalog/view/56/411/946>).

The edition is partially normalised for easier full-text search. This results in a partially normalised edition that retains as many visual features of the text while supporting later NLP processing.

- **Early modern spelling:** U and V are used interchangeably; J is often used for I. Keep J as written, since it can be automatically replaced. For U/V, normalise where clear (e.g. *ut*), but **if unsure, leave unchanged**.
- **Ligatures and abbreviations:** Ligatures such as *AE/ae*, *OE/oe*, or long *ſt* should be normalised. Standard abbreviations (e.g. *q;* for *que*, superscript ⁻⁹ for *us*) should be expanded. An *R* may stand for *rum*, *M* with tilde for *mm*, *N* with tilde for *nn*, *U* with overline for a missing letter (often *um*). **If unsure, leave unchanged** rather than risk introducing an error. **The guiding principle is to remain faithful to what is visible in the text.** Unclear readings should not be (over-)interpreted; expansions can be handled (semi-automatically) later.
- **Diacritics and symbols:** The *NOSCEMUS* model sometimes retains diacritics and special characters (e.g. ampersands), but not consistently (sometimes = &, sometimes = “et”). Do not attempt to harmonise these manually. If a diacritic appears in the output, leave it; if absent, do not add it. The same applies to related symbols (such as &): do not remove them if present, but do not introduce them if missing. This saves time, and diacritics can later be removed automatically with mapping functions. The LLM correction prompt may suggest removing them—ignore this where it conflicts with the rule of preserving what the model produced. Since the main aim is to capture semantic content rather than fine orthographic details, diacritics and minor variants are not a priority.
- **Missing lines:** If a whole line is missing, please add/draw the line and transcribe it in the correct place.

- **All-caps words** → uppercase.
- Pay attention to **missing punctuation and set logical negation signs** for broken-up words at the end of a line. → Insert logical negation signs (where missing) when words are split across lines, as this facilitates later automatic word reunification.
- **Initials:** Ideally, mark initials as regions, but I usually integrate them into the line of the word to which they belong, producing cleaner output. Initials are rare in *Michael Maier*, and not a priority. Maintaining correct reading order remains the key principle.
- In general, keep information that can be removed later if needed.
- Layout information is particularly important, as it often signals semantic divisions (e.g. headings).
- **Gaps/missing or illegible text:** If text is completely illegible, mark it as a **gap** (textual tag) so that the TEI document reflects missing content even without the image.
For many *Michael Maier* texts, better-quality scans exist in other libraries. If a page is badly digitised with extensive gaps, I can search for an alternative version. While early modern editions are not always identical, another witness may provide a more accurate transcription. We initially opted for using these poor scans years ago (even though OCR generally works better with better quality scans) and keep them for cost-efficiency reasons now, but where necessary and possible. we can supplement them with better sources to fill any extensive gaps.
- Measurement signs (e.g. drams) are often misread as numbers (especially “3”); check these carefully. For more info, see table here:
<https://de.wikipedia.org/wiki/Apothekergewicht>
- Greek letters, where present, should be transcribed if possible. For occasional Greek letters, you may use the **Lexilogos Ancient Greek Keyboard** (https://www.lexilogos.com/keyboard/greek_ancient.htm), which provides a visual input method. Learning to type Greek directly from your keyboard is possible but likely unnecessary for this project, as the texts contain very few Greek passages. If you are not confident transcribing Greek, mark the letters clearly or let me know which pages contain them, and I can handle them.
- If you encounter **Ramistic diagrams**, these will mainly appear in recipe books, though they may also occur in *Michael Maier* texts. In Transkribus, you may use regions (including nested regions if needed) to preserve reading order. However, do not spend too much time on detailed encoding: Transkribus cannot fully capture their complexity. Proper encoding would require direct work in TEI. For now, do your best within the tool and leave them as is. They are not a priority for me.

- **Alchemical symbols:** Ideally, transcribe them where possible. Only some are available in Unicode ([https://en.wikipedia.org/wiki/Alchemical_Symbols_\(Unicode_block\)](https://en.wikipedia.org/wiki/Alchemical_Symbols_(Unicode_block))); I maintain a curated list of preferred characters. If multiple similar symbols exist, use the same one consistently. However, mistakes in this area may be unavoidable given the complexity. For **alchemical characters not available in Unicode**, I have sometimes used the **@ symbol** as a placeholder, either for unknown symbols or for cases where the transcriber could not identify the character. This is not ideal for OCR training, since the model would treat them as identical, but it ensures the symbol can later be located and corrected. If you resort to such placeholders outside the recipe texts, please let me know, so that no unknown characters remain unnoticed. Alternatively, you may encode them as gaps.
- **Fine-tuned models:** For recipes, I created a fine-tuned *NOSCEMUS* model re-trained on a few pages of Wecker's *Antidotarium* (5 pages training, 1 validation), which already improved recognition of measurement symbols. This model is available to you in the recipes collection. Alternatively, you may retrain the model with your own corrected pages, though this will likely require credits or a short Transkribus subscription.

Alchemical Special Characters (from Magnet, Pharmacopoeia)

CHARACTERES

ACETUM. ☩

Acetum destilla-
rum. ☩

Aes ustum. ☿

Alembicus. ☿

Alumen. □

Amphora. 🌀

Amalgama. $\overline{aaa}$

Ana. ã an.

Antimonium. ♂

Aqua. ∇

Aqua fortis. ∇

Aqua Regia, Siygià. ∇

Arcitenens. ♂

Arena. ∴

Argentum. ☾

Argentum Vivum. ♀

Aries. ♈

Arsenicum. ☉

Auripigmentum. ☉

Aurum. ☉

Balneum. B.

Balneum Mariae, s. Maris. MB

Balneum Vaporis, s. Roris. VB

Calx. c

Calx viva. ♀

Cancer. ♋

Caper. ♎

Caput mortuum. ☹

Chalybs. ♂

Cineres. ☼

Cinnabaris. ☿

Cornu Cervi. C.C.

Crucibulum. ☩

Cuprum. ♀

Destillare. ↘

Dies. ☽

Drachma. 3

Ferrum. ♂

Gemini. ♊

Hora. ⌘

Ignis. △

Iupiter. ♃

Leo. ♌

Libra. ♎

Libr. ⚖

Luna. ☾

Mars. ♂

Martis limatura. □

Mercurius. ☿

Mercurius sublimatus. ☿ ☿ ☿

Mercurius praecipitatus. ☿ ☿ ☿

Mensis. ☒

Nitrum. ①

Oleum. ☿

Oppositio. ♂

Piscis. ♓

Plumbum. ♄

Pulvis. ☿

Praecipitare. ☿

Quadratus. □

Quinta Essentia. QE

Retorta. ☿

Sal Armoniacum. ✱

Sal commune. ☉

Sal Gemmae. ♂

Sal Nitrum. ☉

Saturnus. ♄

Scorpius. ♏

Scrupulus. ☿

Semis. ♀

Sextilis. ✱

Sigillare Hermetice. S.H.

Sol. ☉

Spiritus. ☿_{SP}

Spiritus Vini. ♀

Stannum. ♄

Stratum superstratum. S.S.S.

Sublimare. ☿

Talcum. X

Taurus. ♉

Trigonus. △

Venus. ♀

Vinum. V.

Virgo. ♍

Virid. Aeris. ⊕

Vitriolum. ⊕

Vitrum. X.X.

Uncia. ℥

Urina. ☐

ERGÄNZENDES

℥ – halbe Dram

℥ – halbe Unze

♁ – Sulfur (Schwefel) / Seele / Entflammbarkeit

♀ – Tartar (Weinstein)

℞ - Tinktur

Alchemie / Astronomie / Astrologie

℥ - dram (Maßeinheit für Gewicht/Volumen) <https://symbl.cc/en/0292/>

℥ – Halbe Dram

⊖ - Skrupel (Maßeinheit, klein)

℥ – halbe Unze

℥ - Unze

℔ - Pfund

♃ - Jupiter / Zinn

☉ - Sonne / Gold

☿ - Merkur / Quecksilber

♀ - Venus / Kupfer

- <https://de.wikipedia.org/wiki/Apothekergewicht>
- [https://en.wikipedia.org/wiki/Alchemical_Symbols_\(Unicode_block\)](https://en.wikipedia.org/wiki/Alchemical_Symbols_(Unicode_block))

- https://unicode.org/charts/nameslist/n_1F700.html
- <https://symbll.cc/en/unicode/blocks/alchemical-symbols/>